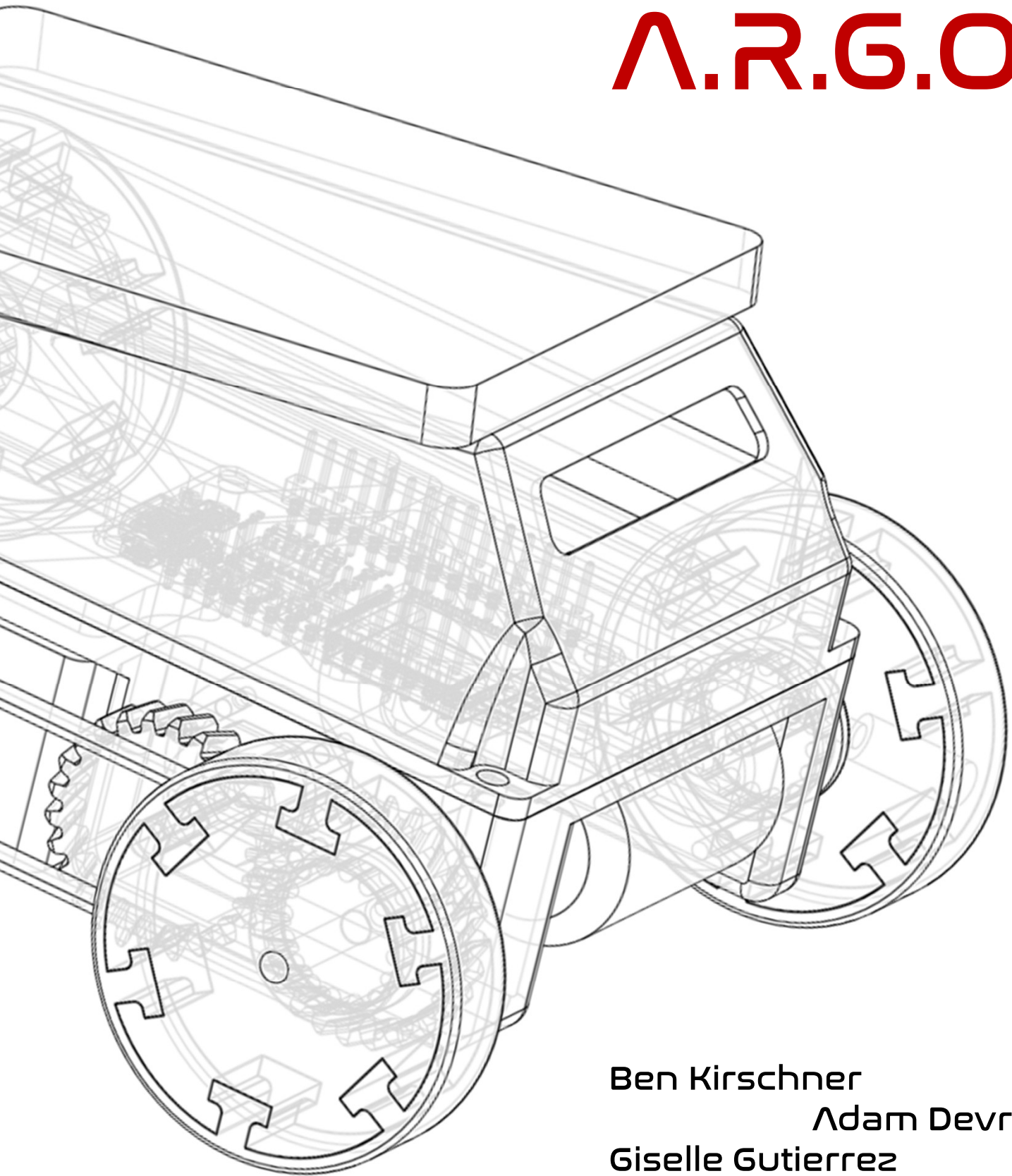


SEC 001

A.R.G.O.



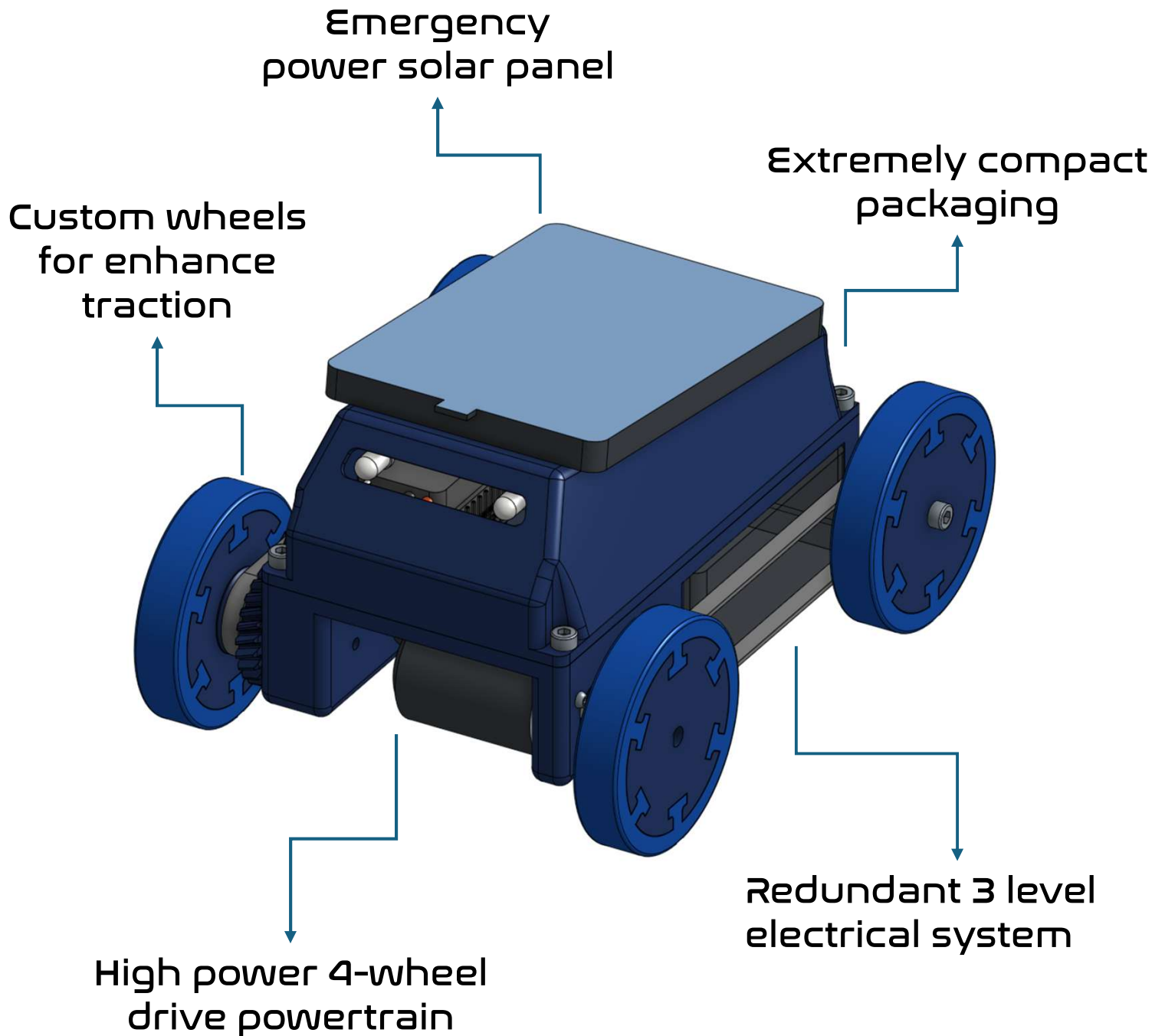
Ben Kirschner

Adam Devries

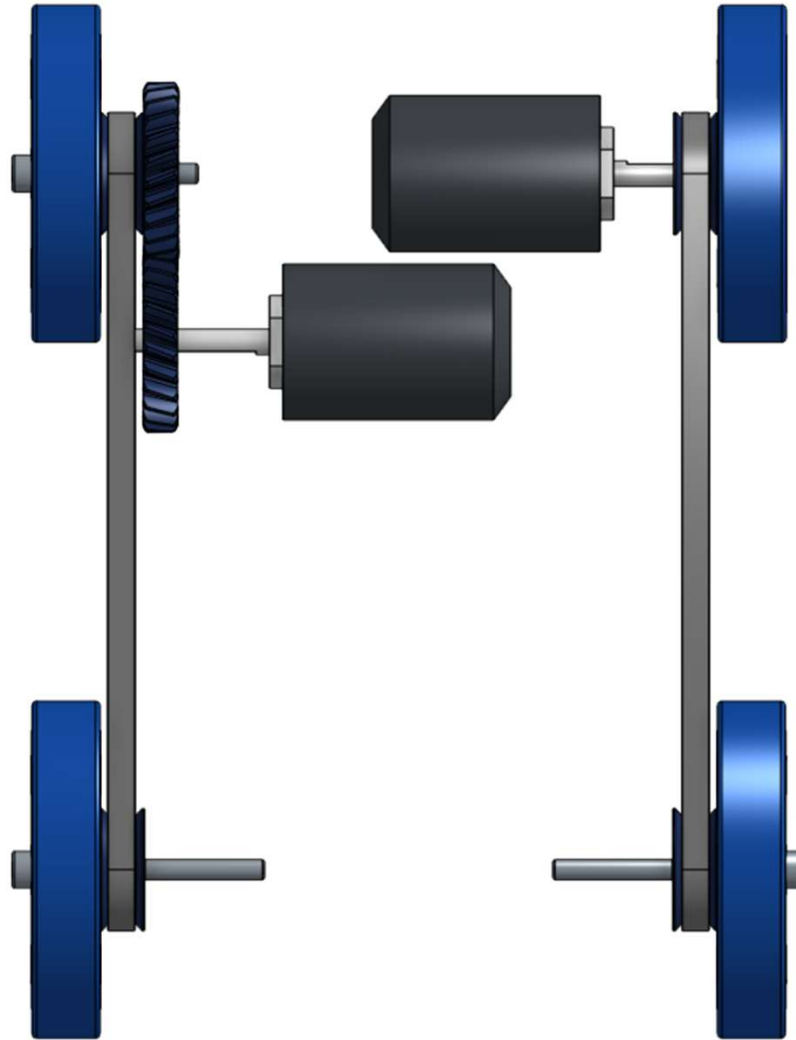
Giselle Gutierrez

Ryan Costanza

DESIGN OVERVIEW



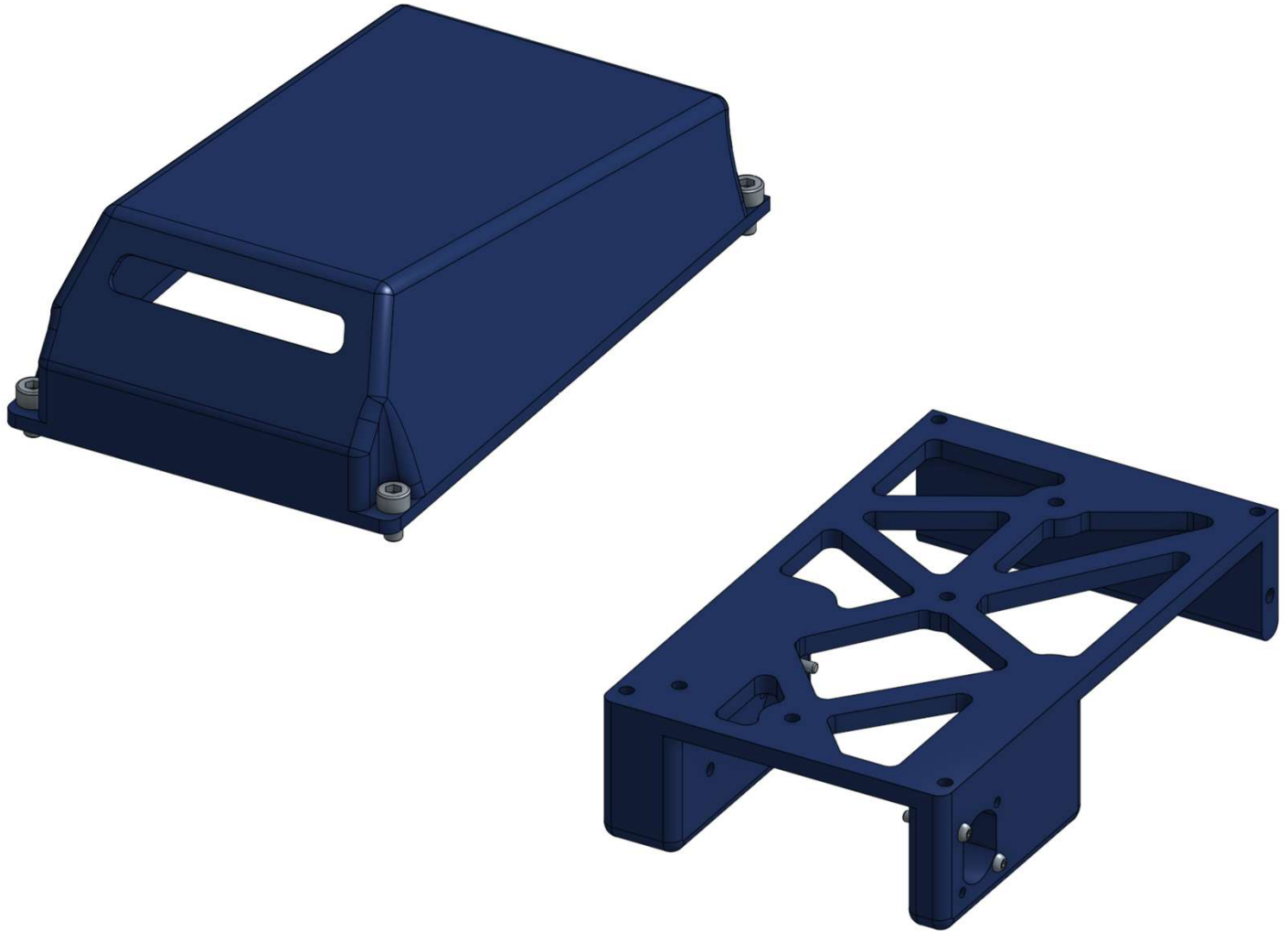
DRIVE SYSTEM



- 2x 2300 KV brushless motors coupled with 22:1 planetary gearboxes provide robust and precise motion
- Custom ordered 88T GT2 belts provide all wheel drive
- Two helical gears allow for drive motor offset, necessitated by tight packaging constraints
- Wheels utilize 2-piece architecture, with TPU fuzzy skin treads to enhance traction

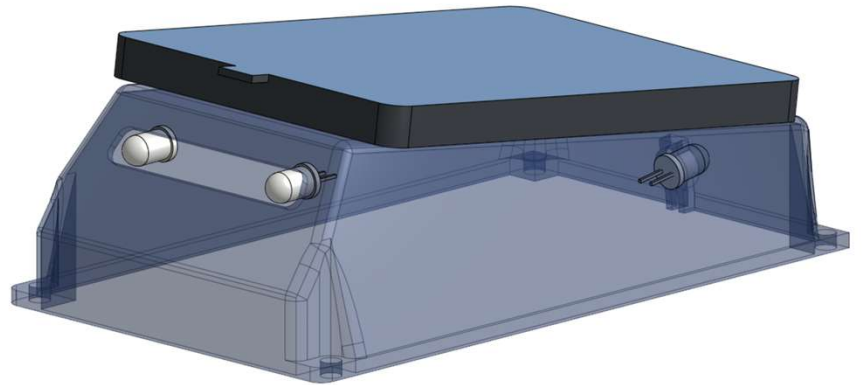
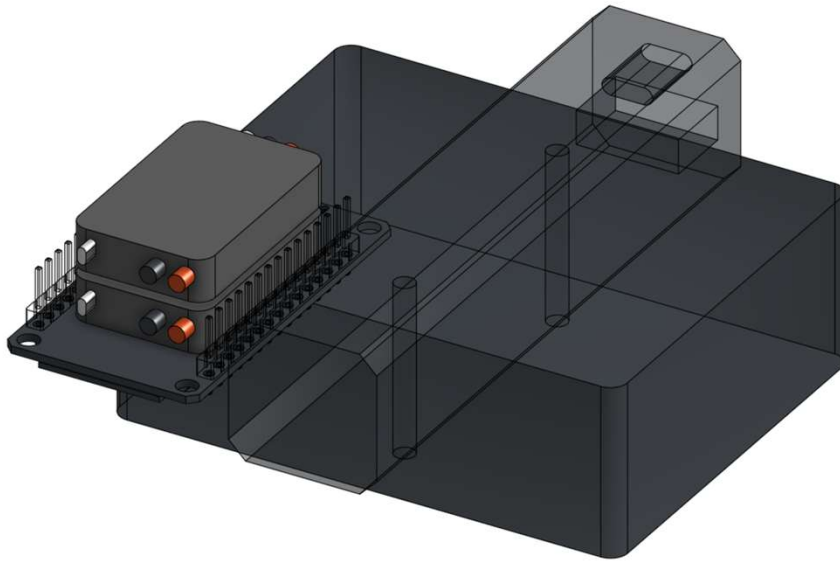
ROVER

STRUCTURE



- Single piece superstructure enhances ease of manufacturing and reduces system complexity
- All structure pieces 3D printed in PLA
- Structure features lightening to balance weight and material use with stiffness under load
- Top shell acts to shield electronics and provide an optimized 8° surface for solar panel mounting

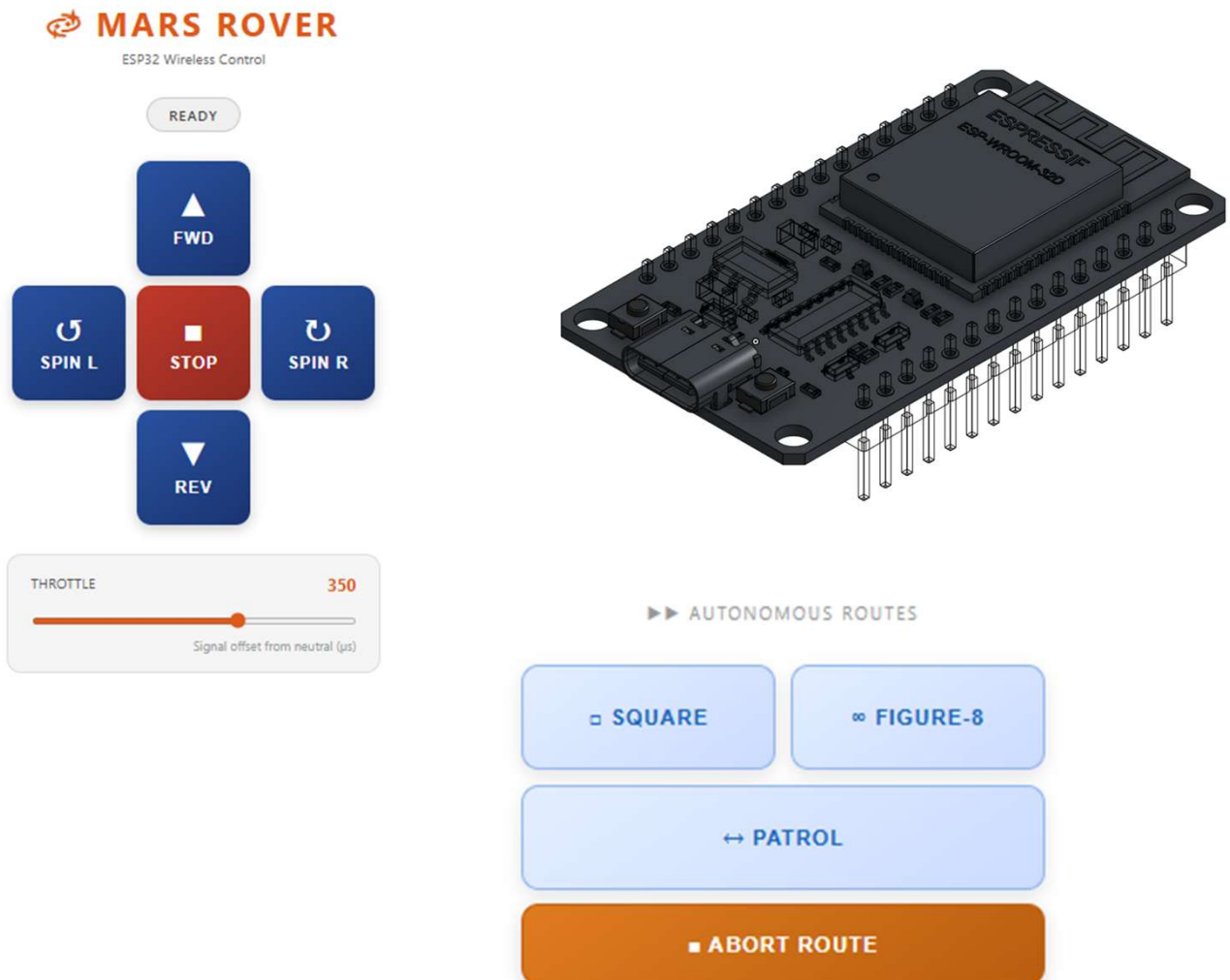
ROVER ELECTRONICS



- Features 3 discrete voltage platforms: 12V drive power, 5V/3.3V AUX power, and 1.3V emergency system
- High capacity 18650 Li-ion cells directly power motors
- Microcontroller and headlights powered from isolated 5V/3.3V circuit to enhance redundancy
- Solar panel and emergency LED beacon are integrated with top shell of the rover structure

ROVER

CONTROL



- ESP-32 Devkit V1 servers as the brains of A.R.G.O. handling drive control and powering the headlights
- Controlled via web server hosted by the microcontroller, allowing any device with Wi-Fi to control the rover
- Program features autonomous route capabilities, able to be created and adjusted directly in the source code
- Drive logic features speed ramping, state control, and direct interfacing with ESC via PWM